

Homework 3

PSTAT 120C: Design and Analysis of Experiments

Instructor: Ali Abuzaid

STUDENT NAME

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Instructions

- This homework aims to familiarize you with real-life data.
- Please write your **full name** and **perm number** legibly on the front page of your homework sheet.
- This homework must be completed and submitted **individually**. Collaboration is allowed for discussion, but each student must submit their own work.
- Solve all homework problems. You may use R for your calculations and analysis, but you must include all R code used.
- Your reasoning must be clear and complete. To receive full credit, provide sufficient explanations alongside any code and results.
- Ensure that all R code, mathematical formulas, and workings are presented clearly and appropriately.
- All figures should be numbered, and axes must be labeled.
- Please use the provided HW 3 S26.qmd file to type your solutions and submit the completed assignment as a PDF file. You can utilize RStudio for this purpose. For guidance, refer to the [Tutorial: Hello, Quarto](#)).
- Submit your solution via **Gradescope** as a **single PDF file**.

Due Date

Due Date: Friday, May 8, 2026 at 11:59 PM

Note to Students

Solve all the questions both manually and using R commands, and ensure that your final answers from both methods are clearly presented and compared.

ANSWER ALL THE FOLLOWING QUESTIONS:

! Question 1

Do you hate Mondays? Researchers in Germany have provided another reason for you: They concluded that the risk of heart attack on a Monday for a working person may be as much as 50% greater than on any other day.¹ The researchers kept track of heart attacks and coronary arrests over a period of 5 years among 330,000 people who lived near Augsburg, Germany. In an attempt to verify the researcher's claim, 200 working people who had recently had heart attacks were surveyed. The day on which their heart attacks occurred appear in the following table.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
24	36	27	26	32	26	29

Do these data present sufficient evidence to indicate that there is a difference in the percentages of heart attacks that occur on different days of the week? Test using $\alpha = 0.05$.

- Obtain the p-value.

ANSWERS TO QUESTION 1:

$$H_0 : p_{Sun} = p_{Mon} = p_{Tue} = p_{Wed} = p_{Thu} = p_{Fri} = p_{Sat} = 1/7$$
$$H_a : \text{at least one } p_i \neq 1/7$$

```
1 n <- 200
2 p <- 1/7
3
4 exp <- n*p
5 obs <- c(24, 36, 27, 26, 32, 26, 29)
6
7 diff <- obs - exp
8 diff
```

```
[1] -4.5714286  7.4285714 -1.5714286 -2.5714286  3.4285714 -2.5714286  0.4285714
```

```
1 chisq <- sum(diff^2)/exp
2 chisq
```

```
[1] 3.63
```

```
1 df <- length(obs) - 1
2
3 # Critical Value
4 crit_val <- qchisq(0.05, df, lower.tail = FALSE)
5 crit_val
```

```
[1] 12.59159
```

```
1 crit_val > chisq # ==> Don't reject H0 if TRUE, Reject H0 if FALSE
```

```
[1] TRUE
```

```
1 # Finding p-value
2 p_val <- pchisq(chisq, df, lower.tail = FALSE)
3 p_val
```

```
[1] 0.7266011
```

At a significance level $\alpha = 0.05$, we do not have enough evidence to conclude that there is a difference in the percentages of heart attacks that occur on different days of the week.

0.1 Written Work

```
1 knitr::include_graphics('HW3Q1.jpeg')
```

HW3

Q1. $n = 200$, $\alpha = 0.05$

Day	Obs	Exp	(Obs - Exp) ² / Exp
Sun	24	28.67	-4.671
Mon	36	28.67	7.328
Tues	27	28.67	-1.671
Wed	26	28.67	-2.671
Thurs	32	28.67	3.328
Fri	26	28.67	-2.671
Sat	29	28.67	0.028
Total	200	200	3.62

$E_{xp} = n \times p = 200 \times 1/7 = 28.57$

$H_0: p_S = p_M = p_T = p_W = p_F = p_{Sa} = 1/7$

$H_a: \text{at least one } p_i \neq 1/7$

Test Statistic: $\chi^2 = \sum_{i=1}^7 \frac{(Obs_i - Exp_i)^2}{Exp_i} = 3.63$

RR: If $\chi^2_{obs} > \chi^2_{0.05, 6}$, Reject H_0

Since $\chi^2_{obs} = 3.63 < 12.59159$, then we do not reject H_0 .
 That is we do not have enough evidence to conclude

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! Question 2

The data in the following table are the frequency counts for 400 observations on the number of bacterial colonies within the field of a microscope, using samples of milk film.² Is there sufficient evidence to claim that the data do not fit the Poisson distribution? (Use $\alpha = 0.05$.)

Note: For the R solution, you may use any appropriate built-in function where applicable, **and also** provide your own code to perform the calculations.

Number of Colonies per Field	Frequency of Observation
0	56
1	104
2	80
3	62
4	42
5	27
6	9
7	9
8	5
9	3
10	2
11	0
19	1
400	

ANSWERS TO QUESTION 2:

H_0 : Data follows a Poisson Distribution

H_a : Data DOES NOT follow a Poisson Distribution

```
1 # Inputting Data
2 counts <- c(rep(0, 56), rep(1, 104), rep(2, 80), rep(3, 62),
3             rep(4, 42), rep(5, 27), rep(6, 9), rep(7, 9), rep(8, 5),
4             rep(9, 3), rep(10, 2), rep(19, 1))
5
6 n <- length(counts)
7
8 freq <- table(counts)
9
10 obs_freq <- c(freq["0"], freq["1"], freq["2"], freq["3"], freq["4"],
11              freq["5"], freq["6"], sum(freq[as.numeric(names(freq)) >= 7]))
12
13 result <- data.frame(
14   Yi = c("0", "1", "2", "3", "4", "5", "6", "7+"),
15   Frequency = as.numeric(obs_freq))
16
17 result
```

	Yi	Frequency
1	0	56

2	1	104
3	2	80
4	3	62
5	4	42
6	5	27
7	6	9
8	7+	20

```

1 # Lambda MLE
2 lambda_MLE <- sum(counts) / n
3
4 # Estimating Frequencies Under H0 (using Lambda MLE) (Must be larger than 5)
5 exp0 <- dpois(0, lambda_MLE) * n
6 exp1 <- dpois(1, lambda_MLE) * n
7 exp2 <- dpois(2, lambda_MLE) * n
8 exp3 <- dpois(3, lambda_MLE) * n
9 exp4 <- dpois(4, lambda_MLE) * n
10 exp5 <- dpois(5, lambda_MLE) * n
11 exp6 <- dpois(6, lambda_MLE) * n
12 exp7plus <- n - (exp0 + exp1 + exp2 + exp3 + exp4 + exp5 + exp6)
13
14 exp_freq <- c(exp0, exp1, exp2, exp3, exp4, exp5, exp6, exp7plus)
15 exp_freq

```

```

[1] 34.864341 85.068991 103.784169 84.411124 51.490786 25.127503 10.218518
[8] 5.034568

```

```

1 # Pearson ChiSq Test
2
3 chisq2 <- sum((obs_freq - exp_freq)^2 / exp_freq)
4 chisq2

```

```

[1] 74.94605

```

```

1 df2 <- length(obs_freq) - 1 - 1 # lost additional df for estimating parameter lambda)
2
3 p_val2 <- pchisq(chisq2, df2, lower.tail = FALSE)
4 p_val2

```

```

[1] 3.937753e-14

```

```

1 # Critical Value
2
3 crit_val2 <- qchisq(0.05, df2, lower.tail = FALSE)
4 crit_val2 > chisq2 # TRUE ==> Fail to Reject H0, FALSE ==> Reject H0

```

```

[1] FALSE

```

crit_val2

[1] 12.59159

Since our p-value is basically 0, at the significance level $\alpha = 0.05$, we will reject the null hypothesis, that is, we have sufficient evidence to claim that our data DOES NOT follow a Poisson Distribution.

0.2 Written Work

knitr::include_graphics('HW3Q2.jpeg')

Date: Q2:

H_0 : Data follows a Poisson distribution
 H_a : Data does not follow a Poisson distribution

- We need cells w/ 5 or expected counts is, as such we will define the cells as follows:
 $Y=0, Y=1, Y=2, Y=3, Y=4, Y=5, Y=6, Y \geq 7$

$P(Y|X) = \frac{e^{-\lambda} \lambda^Y}{Y!}$, $\lambda_{MLE} = \bar{Y} = \frac{\sum Y_i}{n} = \frac{(0 \cdot 56) + (1 \cdot 104) + \dots}{400} = 2.44$

$\hat{p}_0 = P(Y=0|\lambda_{MLE}) = e^{-\lambda}$
 $\hat{p}_1 = P(Y=1|\lambda_{MLE}) = \lambda e^{-\lambda}$
 $\vdots$
 $P(Y \geq 7|\lambda_{MLE}) = 1 - \hat{p}_0 - \hat{p}_1 - \hat{p}_2 - \hat{p}_3 - \hat{p}_4 - \hat{p}_5 - \hat{p}_6$

$E[n_i] = n \cdot \hat{p}_i$

Test Statistic: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

$= \frac{(56-34.88)^2}{34.88} + \frac{(104-85.04)^2}{85.04} + \dots + \frac{(20-5.0368)^2}{5.0368} = 24.94605$

$\chi^2_{0.05, 458} = 12.59159$

Since $24.94605 > 12.59159$, then we reject H_0 .
is we have sufficient evidence to say that our data does not follow a Poisson Distribution.

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! Question 3

Does education really make a difference in how much money you will earn? Researchers randomly selected 100 people from each of three income categories—"marginally rich," "comfortably rich," and "super rich"—and recorded their education levels. The data is summarized in the table that follows.

Highest Education Level	Marginally Rich	Comfortably Rich	Super Rich
No college	32	20	23
Some college	13	16	1
Undergraduate degree	43	51	60
Postgraduate study	12	13	16
Total	100	100	100

1. Describe the independent multinomial populations whose proportions are compared in the χ^2 analysis.
2. Do the data indicate that the proportions in the various education levels differ for the three income categories? Test at the $\alpha = 0.01$ level.

ANSWERS TO QUESTION 3:

P1:

Independent Populations - The 3 distinct groups researchers chose to sample, in this case, the Income Categories: Marginally Rich, Comfortably Rich, Super Rich

Multinomial Populations: The 4 mutually exclusive education levels people fall into within the 3 independent populations, the analysis will test whether there is a difference in the education level proportions across the 3 populations: No College, Some College, Undergraduate Degree, Postgraduate Study

P2:

H_0 : The proportion of the 4 education levels is the same across the 3 income categories (Homogenous)

H_a : The proportion in the 3 education levels differ across the 3 income categories (Not Homogenous)

```
1 data <- matrix(c(32, 20, 23,
2                 13, 16, 1,
3                 43, 51, 60,
4                 12, 13, 16),
5                 nrow = 4, byrow = TRUE)
6
7 colnames(data) <- c("Marginally_Rich", "Comfortably_Rich", "Super_Rich")
8 rownames(data) <- c("No_College", "Some_College", "Undergraduate", "Postgraduate")
9
10 data
```

	Marginally_Rich	Comfortably_Rich	Super_Rich
No_College	32	20	23
Some_College	13	16	1
Undergraduate	43	51	60
Postgraduate	12	13	16

```

1 # Row Totals
2 total_row1 <- sum(data[1,])
3 total_row2 <- sum(data[2,])
4 total_row3 <- sum(data[3,])
5 total_row4 <- sum(data[4,])
6
7 col_total <- 100 # since sampled 100 people from each income level
8
9 grand_total <- 300
10
11 # Expected Values
12
13 exp_row1 <- (total_row1 * col_total) / grand_total
14 exp_row2 <- (total_row2 * col_total) / grand_total
15 exp_row3 <- (total_row3 * col_total) / grand_total
16 exp_row4 <- (total_row4 * col_total) / grand_total
17
18 # Test Statistic
19 chisq3 <- sum((data[1,] - exp_row1)^2 / exp_row1) + sum((data[2,] - exp_row2)^2 / exp_row2) +
20   sum((data[3,] - exp_row3)^2 / exp_row3) + sum((data[4,] - exp_row4)^2 / exp_row4)
21 chisq3

```

```
[1] 19.17233
```

```

1 df3 <- (nrow(data) - 1)*(ncol(data) - 1)
2
3 # P-value and Critical Value
4
5 p_val3 <- pchisq(chisq3, df3, lower.tail = FALSE)
6 p_val3

```

```
[1] 0.003882283
```

```

1 crit_val3 <- qchisq(0.01, df3, lower.tail = FALSE)
2 crit_val3 > chisq3 # TRUE ==> Fail to Reject H0, FALSE ==> Reject H0

```

```
[1] FALSE
```

```
1 crit_val3
```

```
[1] 16.81189
```

At the significance level $\alpha = 0.01$, our critical value is 16.81189 which is smaller than our observed test statistic of 19.172, which means we are within the rejection region and thus, will reject H_0 . That is, we have sufficient evidence to say that there is a difference in the education level proportions across the 3 income categories.

0.3 Written Work

```
1 knitr::include_graphics('HW3Q3.jpeg')
```

Q3: H_0 : Proportion of 4 education levels is the same across income levels
 H_a : Proportion of 4 education levels differs across income levels

Date: _____

$$E_{AC,MR} = \frac{26 \times 100}{300}, E_{AC,ER} = \frac{75 \times 100}{300}, E_{AC,SE} = \frac{75 \times 100}{300} = 25$$

$$E_{SC,MR} = \frac{30 \times 100}{300}, E_{SC,ER} = \frac{30 \times 100}{300}, E_{SC,SE} = \frac{30 \times 100}{300} = 10$$

$$E_{UC,MR} = \frac{154 \times 100}{300}, E_{UC,ER} = \frac{154 \times 100}{300}, E_{UC,SE} = \frac{154 \times 100}{300} = 51.33$$

$$E_{PS,MR} = \frac{41 \times 100}{300}, E_{PS,ER} = \frac{41 \times 100}{300}, E_{PS,SE} = \frac{41 \times 100}{300} = 13.667$$

Test Statistic: $\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$

$$= \frac{(32-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(23-25)^2}{25} + \frac{(13-10)^2}{10} + \frac{(46-10)^2}{10} + \dots$$

$$+ \frac{(13-13.667)^2}{13.667} + \frac{(14-13.667)^2}{13.667} = 19.172$$

$$\chi^2_{0.01, df=4-1)(3-1)=6} = 16.8119$$

Since $19.172 > 16.8119$, we reject H_0 , that is we have sufficient evidence to say there is a difference in education level across the 3 income categories.

0.3.1 Verification

```
1 chisq.test(data)
```

Pearson's Chi-squared test

data: data

X-squared = 19.172, df = 6, p-value = 0.003882

! Question 4

The results of a study suggest that the initial electrocardiogram (ECG) of a suspected heart attack victim can be used to predict in-hospital complications of an acute nature. The study included 469 patients with suspected myocardial infarction (heart attack). Each patient was categorized according to whether their initial ECG was positive or negative and whether the person suffered life-threatening complications subsequently in the hospital. The results are summarized in the following table.

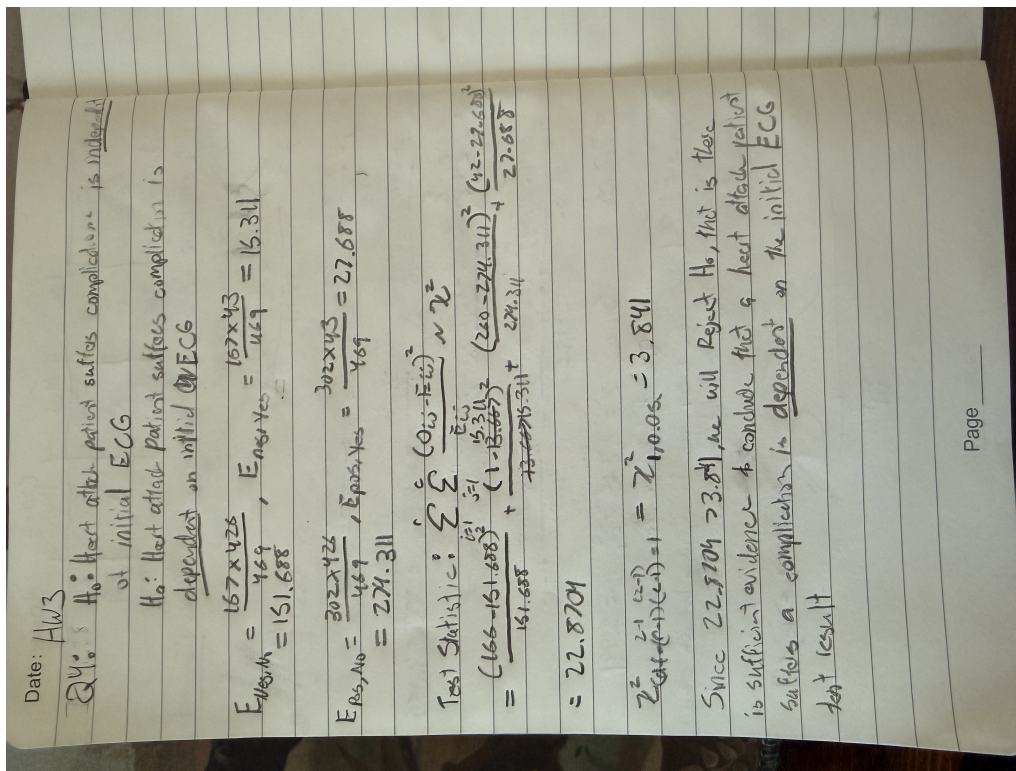
ECG	Subsequent In-Hospital Life-Threatening Complications		Total
	No	Yes	
Negative	166	1	167
Positive	260	42	302
Total	426	43	469

Is there sufficient evidence to indicate that whether or not a heart attack patient suffers complications depends on the outcome of the initial ECG? Test using $\alpha = 0.05$.

ANSWERS TO QUESTION 4:

0.3.2 Written Work

```
1 knitr::include_graphics('HW3Q4.jpeg')
```



0.3.3 Using R functions

```
1 data2 <- matrix(c(166, 1,  
2                  260, 42),  
3                  nrow = 2, byrow = TRUE)  
4  
5 colnames(data2) <- c("No", "Yes")  
6 rownames(data2) <- c("Negative", "Positive")  
7  
8 chisq.test(data2)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: data2  
X-squared = 21.3, df = 1, p-value = 3.927e-06
```